

From Threshold Belief to Epistemic Phase Geometry: Mechanized Dynamics in a Four-Valued Probabilistic Epistemic Logic

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Abstract

I study the dynamics of threshold belief and evidence-stable knowledge in a finite four-valued probabilistic epistemic logic based on Belnap–Dunn/FDE values. Positive and negative support are tracked independently, so each formula occupies one of four complete statuses: truth only, falsity only, both, or neither. Probabilistic belief thresholds the positive and negative support masses separately, while evidence-stable knowledge filters that belief through a qualitative invariance test on the complete accessible FDE profile. The resulting factorization is exact: knowledge equals threshold belief on stable profiles and collapses to strict falsity on unstable profiles.

The dynamic results develop this architecture from endpoint reachability to an explicit phase geometry. Admissible conditionalization changes only local probability. A six-world family realizes every ordered transition between the four FDE values at the level of $K(\mathbf{B}p)$, so the categorical reachability graph is complete. At the same time, belief is pointwise invariant exactly when neither of its two Lockean threshold bits changes. This yields a Boolean threshold square with a Hamming-style wall count $d_{\text{thr}} \in \{0, 1, 2\}$. A categorical wall flip is equivalent to endpoint straddling of the corresponding rational support mass. For the directed affine interpolation $\gamma(t) = x + t(y - x)$, straddling yields an explicit rational threshold-hit parameter in $[0, 1]$. Two-wall transitions therefore carry two unique crossing times. Their order is positive-first, simultaneous, or negative-first; simultaneity is exactly passage through (c, c) in the two-dimensional support plane. If the crossings are sequential, the rational midpoint between them occupies a necessary adjacent FDE phase. Thus a diagonal endpoint transition such as $\mathbf{N} \rightarrow \mathbf{B}$ decomposes along the affine support path as either $\mathbf{N} \rightarrow \mathbf{T} \rightarrow \mathbf{B}$ or $\mathbf{N} \rightarrow \mathbf{F} \rightarrow \mathbf{B}$, according to crossing order.

The mechanization now also supplies a stronger finite-probability layer. Nonnegative normalized rational world weights generate finitely additive, extensional local measures; rational convex interpolation preserves valid distributions; and every fixed event mass varies affinely along the same parameter. These local paths lift to complete **StrongProbability Model** paths with worlds, accessibility, valuation, and threshold fixed. Thus model-valued realizability is verified for weight-generated endpoints on a common semantic skeleton. The remaining boundary is formula-level and update-level: probability-sensitive formulas can change the event whose mass is measured, and the intermediate models are not claimed to arise from repeated conditionalization of the source. No general continuity or measure-theoretic theorem is claimed. A systematic literature audit is also still required before any strong novelty claim.

1 Introduction

Probabilistic epistemic models are usually asked two different questions. One concerns degrees: how much support does an agent assign to a proposition? The other concerns categories: does the agent believe, know, reject, or remain undecided? Threshold approaches connect the two by turning sufficiently high probability into all-or-nothing belief. Four-valued semantics adds a

second axis. Positive and negative support can coexist or both fail, so the categorical result is no longer binary but belongs to

$$\{\mathbf{T}, \mathbf{F}, \mathbf{B}, \mathbf{N}\} = \{(1, 0), (0, 1), (1, 1), (0, 0)\},$$

following the Belnap–Dunn/FDE tradition [1, 2].

The 4-PEL project combines these ideas in a finite probabilistic epistemic semantics. Positive and negative extensions are assigned local probability masses independently. A Lockean threshold is applied to each mass, producing a four-valued belief state. A separate evidence-stable knowledge operator then checks whether the complete FDE profile is invariant across the accessible range. This yields an exact factorization:

$$\mathbf{K}_i\varphi = \begin{cases} \mathbf{B}_i\varphi, & \text{Stable}_i(\varphi), \\ \mathbf{F}, & \text{otherwise.} \end{cases}$$

The knowledge operator is therefore not a second probability threshold. It is a qualitative stability filter on top of threshold belief.

The present paper asks what happens dynamically when local probability is updated while accessibility, atomic valuation, and threshold remain fixed. The central research question is:

Research Question 1.1 (Dynamic phase geometry). How does admissible probabilistic conditionalization move four-valued belief and knowledge between $\mathbf{T}, \mathbf{F}, \mathbf{B}, \mathbf{N}$, and what geometric and temporal structure is hidden by the endpoint transition alone?

The machine-checked answer develops in several layers. First, admissible conditionalization can both destroy and restore the stability gate. More strongly, a fixed six-world construction scheme realizes every ordered pair of four-valued values at the level of $\mathbf{K}(\mathbf{B}p)$. Thus the categorical reachability graph is complete. That global permissiveness is paired with a sharp local invariant: a belief value changes exactly when at least one of its two threshold decisions changes.

The four categorical states can consequently be treated as the vertices of a Boolean square. The number of changed threshold coordinates defines a wall count

$$d_{\text{thr}} \in \{0, 1, 2\}.$$

Zero walls means the same FDE state, one wall means an edge move, and two walls mean a diagonal move. The two diagonals are $\mathbf{T} \leftrightarrow \mathbf{F}$ and $\mathbf{N} \leftrightarrow \mathbf{B}$.

The geometric interpretation can then be reconnected to the underlying rational support masses. A changed threshold bit is exactly endpoint straddling of the Lockean boundary. For the explicit affine interpolation

$$\gamma(t) = x + t(y - x), \quad t \in \mathbb{Q},$$

straddling produces a rational crossing parameter $t \in [0, 1]$ at which $\gamma(t) = c$. A two-wall transition therefore contains two threshold events, one for positive support and one for negative support. Each affine crossing time is unique. Their order is positive-first, simultaneous, or negative-first, and the simultaneous case is exactly passage through the point (c, c) in the support plane.

If the two crossings are not simultaneous, the rational midpoint between them occupies a categorical state adjacent to both diagonal endpoints. The order of the wall events determines the intermediate state. For example,

$$\mathbf{N} \rightarrow \mathbf{B}$$

becomes either

$$\mathbf{N} \rightarrow \mathbf{T} \rightarrow \mathbf{B}$$

or

$$\mathbf{N} \rightarrow \mathbf{F} \rightarrow \mathbf{B}.$$

The same classification holds for the other diagonal orientations.

A further verified layer now connects this geometry to complete probabilistic models. Non-negative normalized rational point weights generate local measures satisfying an explicit finite-probability integrity contract. Convex interpolation preserves those distributions and makes the mass of every fixed event affine in the interpolation parameter. The local paths can then be packaged into complete strong 4-PEL models while worlds, accessibility, atomic valuation, and threshold remain fixed.

This yields a compact structural picture:

$ \begin{aligned} &\text{probability simplex} \longrightarrow \text{strong model path} \\ &\longrightarrow \text{affine fixed-event masses} \longrightarrow \text{threshold sides} \\ &\quad \longrightarrow \text{wall crossings} \longrightarrow \text{crossing order} \\ &\quad \longrightarrow \text{FDE phase sequence.} \end{aligned} $

Methodological discipline. The paper distinguishes four levels throughout. Machine-checked theorems are reported as such. Finite witness constructions are not silently generalized beyond their quantified statement. Geometric and philosophical readings are identified as interpretations. Literature and novelty claims remain deliberately modest until a systematic comparison has been completed.

Current boundary. Model-valued convex paths are now verified for weight-generated strong endpoint models on a common semantic skeleton. This does not yet imply that every legacy conditionalization witness has such a strong representation, that every intermediate point is itself generated by conditionalization of the source model, or that support events defined by arbitrary probability-sensitive modal formulas remain fixed along the path. The rational affine crossing theorem also remains distinct from an abstract intermediate-value theorem over arbitrary continuous paths. These formula-level, update-level, and topological questions are the next mathematical boundaries.

2 Four-Valued Probabilistic Epistemic Semantics

The formal development uses a finite Kripke-style model

$$M = (W, R, \mu, v, c),$$

where W is a finite list of worlds, $R_i(w)$ is the finite accessible range for agent i at world w , $\mu_{i,w}$ is a rational local set function, v assigns complete FDE values to atoms, and c_i is a Lockean threshold satisfying

$$\frac{1}{2} < c_i \leq 1.$$

The model interface requires normalization on the accessible range and on the empty set.

2.1 Bilateral four-valued information

A semantic value is a pair of Boolean coordinates

$$x = (x^+, x^-),$$

with the four canonical values

$$\mathbf{T} = (1, 0), \quad \mathbf{F} = (0, 1), \quad \mathbf{B} = (1, 1), \quad \mathbf{N} = (0, 0).$$

The positive coordinate records support for the formula and the negative coordinate records support against it. Thus $\mathbf{B}$ is a glut and $\mathbf{N}$ is a gap. Negation exchanges the two coordinates, while conjunction takes conjunction on positive support and disjunction on negative support.

This bilateral representation is crucial for the dynamic analysis. The two coordinates can cross a probabilistic threshold at different times, producing a temporal structure that would be invisible in a single Boolean belief status.

2.2 Threshold belief

For a modal formula φ , define its positive and negative accessible extensions at (i, w) by

$$E_{i,w}^+(\varphi) = \{u \in R_i(w) : \llbracket \varphi \rrbracket_u^+ = 1\},$$

$$E_{i,w}^-(\varphi) = \{u \in R_i(w) : \llbracket \varphi \rrbracket_u^- = 1\}.$$

The corresponding support masses are

$$P_{i,w}^+(\varphi) = \mu_{i,w}(E_{i,w}^+(\varphi)), \quad P_{i,w}^-(\varphi) = \mu_{i,w}(E_{i,w}^-(\varphi)).$$

Probabilistic belief thresholds the two masses independently:

$$\mathbf{B}_i\varphi = \left([P_{i,w}^+(\varphi) \geq c_i], [P_{i,w}^-(\varphi) \geq c_i] \right).$$

Accordingly, $\mathbf{B}_i\varphi = \mathbf{B}$ is possible when both signed support masses cross the threshold, while $\mathbf{B}_i\varphi = \mathbf{N}$ occurs when neither does.

2.3 Evidence-stable knowledge

The modal object language contains primitive probabilistic belief $\mathbf{B}$, primitive evidence-stable knowledge $\mathbf{K}$, and primitive raw accessibility possibility. The latter is kept distinct from the internal expression $\neg\mathbf{K}\neg\varphi$, because the two need not agree in a four-valued setting.

For the present paper, the central modal predicate is full-value stability. Let

$$\text{Stable}_i^M(w, \varphi)$$

mean that every world in $R_i(w)$ gives φ the same complete FDE value. The formal knowledge semantics treats heterogeneity itself as negative evidence against knowledge.

The resulting relationship between knowledge and belief is exact.

Theorem 2.1 (Knowledge as stability-filtered belief). *For every finite 4-PEL model, agent, world, and modal formula,*

$$\mathbf{K}_i\varphi = \begin{cases} \mathbf{B}_i\varphi, & \text{Stable}_i(\varphi), \\ \mathbf{F}, & \neg\text{Stable}_i(\varphi). \end{cases}$$

STATUS: LEAN-VERIFIED.

Hence an unstable accessible profile is not merely low-confidence evidence. It is a structural failure mode that absorbs every probabilistic belief value into strict $\mathbf{F}$ at the knowledge layer. Conversely, on a homogeneous profile the stability filter is transparent and $\mathbf{K}_i\varphi$ preserves the complete four-valued belief state.

This factorization will later separate two dynamic channels: probability can move the threshold belief value, and updates can also move a formula between stable and unstable profile phases.

3 Admissible Conditionalization and Dynamic Knowledge Phases

The update operator conditions each local probability function on the positive extension of an evidence formula E . Let

$$E_{i,w}^+ = \{u \in R_i(w) : \llbracket E \rrbracket_u^+ = 1\}.$$

For a finite set $S \subseteq W$, the raw updated local probability is

$$\mu_{i,w}^E(S) = \frac{\mu_{i,w}(S \cap E_{i,w}^+)}{\mu_{i,w}(E_{i,w}^+)}$$

whenever the denominator is nonzero. At the model level, the Lean constructor requires an explicit *ConditionalizationAdmissible* proof. This supplies positive evidence mass at every local state together with the normalization obligations of the updated measure.

The update is intentionally conservative:

$$W^E = W, \quad R^E = R, \quad v^E = v, \quad c^E = c,$$

and only μ changes. Thus any dynamic change in a formula must ultimately be mediated by a semantic dependency on probability.

3.1 A syntactic invariant sector

The modal language has a probability-free fragment containing atoms, negation, conjunction, primitive knowledge, and raw possibility, but no occurrence of the probabilistic belief constructor. Lean proves that every such formula preserves its complete FDE value under admissible conditionalization. Its accessible stability profile, outer knowledge, and raw possibility are therefore preserved as well.

This result isolates a first dynamic boundary:

$$\text{no probabilistic belief occurrence} \implies \text{complete value invariant under conditionalization.}$$

The converse is false. Later sections exhibit formulas containing $\mathbf{B}$ whose probabilities change while their threshold status remains invariant.

3.2 Two dynamic channels for knowledge

The factorization theorem from Section 2 yields a complete two-state classification for $\mathbf{K}_i\varphi$. Let M_0 and M_1 be two models on the same modal language. The stability states of φ determine how the belief channel is exposed at the knowledge layer:

Stable $_{M_0}$	Stable $_{M_1}$	knowledge phase
true	true	$\mathbf{K} = \mathbf{B}$ on both sides
true	false	posterior $\mathbf{K} = \mathbf{F}$
false	true	prior $\mathbf{K} = \mathbf{F}$, posterior $\mathbf{K} = \mathbf{B}$
false	false	$\mathbf{K} = \mathbf{F}$ on both sides

The classification is update-independent: it follows directly from the pointwise factorization. The conditionalization development additionally contains finite witnesses for both off-diagonal phases. One update fractures a homogeneous $\mathbf{B}p$ profile into a heterogeneous profile, giving

$$\text{Stable}(\mathbf{B}p) : \text{true} \rightarrow \text{false}, \quad \mathbf{K}(\mathbf{B}p) : \mathbf{T} \rightarrow \mathbf{F}.$$

A second witness restores stability in the opposite direction:

$$\text{Stable}(\mathbf{B}p) : \text{false} \rightarrow \text{true}, \quad \mathbf{K}(\mathbf{B}p) : \mathbf{F} \rightarrow \mathbf{T}.$$

Corollary 3.1 (No monotone knowledge direction). *Admissible probabilistic conditionalization is neither intrinsically monotone knowledge gain nor intrinsically monotone knowledge loss in the evidence-stable semantics. STATUS: LEAN-VERIFIED.*

This phase table motivates the stronger question addressed next. Instead of asking only whether stability can fracture or recover, one can ask which complete four-valued endpoints are reachable when the belief profile is kept homogeneous so that the outer knowledge filter remains transparent.

4 Complete Dynamic Epistemic Reachability

The strongest endpoint theorem uses a uniform six-world construction scheme. For arbitrary source and target values

$$s, t \in \{\mathbf{T}, \mathbf{F}, \mathbf{B}, \mathbf{N}\},$$

consider worlds

$$\{f, a_1, a_2, a_3, a_4, r\}.$$

The distinguished focus world f carries the target value t for atom p , each of the four anchor worlds a_k carries the source value s , and the spare world r carries $\mathbf{N}$. A second atom e is $\mathbf{T}$ only at f and $\mathbf{F}$ elsewhere. Every world accesses all six worlds. The prior measure is uniform,

$$\mu(\{w\}) = \frac{1}{6},$$

and the Lockean threshold is

$$c = \frac{2}{3}.$$

Before update, the four anchors contribute total mass $4/6 = 2/3$. Coordinate by coordinate, they therefore force the threshold belief $\mathbf{B}p$ to reproduce the source value s , regardless of the single target-bearing focus world. After conditioning on e , all posterior mass is concentrated at f , so $\mathbf{B}p$ reproduces the target value t .

The local probability function is independent of the evaluation world. Consequently, the complete $\mathbf{B}p$ profile is homogeneous across the common accessible range before and after update. The outer knowledge operator therefore passes both values through unchanged.

Theorem 4.1 (Complete dynamic epistemic reachability). *For every ordered pair (s, t) of complete FDE values, there is a member of the six-world family and an admissible conditionalization such that*

$$\mathbf{K}(\mathbf{B}p) : s \longrightarrow t.$$

Across that update the accessibility relation, atomic valuation, and threshold are fixed; only the local probability field changes. STATUS: LEAN-VERIFIED.

Equivalently, the categorical reachability graph is complete:

source \ target	$\mathbf{T}$	$\mathbf{F}$	$\mathbf{B}$	$\mathbf{N}$
$\mathbf{T}$	✓	✓	✓	✓
$\mathbf{F}$	✓	✓	✓	✓
$\mathbf{B}$	✓	✓	✓	✓
$\mathbf{N}$	✓	✓	✓	✓

The theorem immediately gives creation and removal of both nonclassical statuses. In particular, the development contains verified instances

$$\mathbf{T} \rightarrow \mathbf{B}, \quad \mathbf{B} \rightarrow \mathbf{N}, \quad \mathbf{N} \rightarrow \mathbf{T}.$$

Thus neither gluttony nor gappy knowledge is dynamically terminal under the present update semantics.

Quantifier discipline. The theorem does not say that one fixed concrete model realizes all sixteen arrows. The family is parameterized by the requested source and target values. For each pair, however, the pre-update and post-update models share the same R , v , and c , and the transition is generated by admissible probability-only conditionalization. The result is therefore a semantic non-prohibition theorem: no ordered pair of complete four-valued knowledge statuses is ruled out merely by the conditionalization mechanism.

The reachability theorem establishes global permissiveness. The next result identifies the exact local condition under which no categorical change occurs.

5 Threshold-Side Robustness and the Boolean Square

Complete reachability shows how permissive the dynamics can be across different models in the witness family. The complementary question is local: what must change in order for one probabilistic belief value to change at all?

Let

$$p^+ = P_{i,w}^+(\varphi), \quad p^- = P_{i,w}^-(\varphi).$$

Then the complete belief value is exactly the pair of threshold decisions

$$\mathbf{B}_i\varphi = ([p^+ \geq c_i], [p^- \geq c_i]).$$

This gives an immediate semantic boundary.

Theorem 5.1 (Threshold-side robustness). *For any two models on the same language, at a fixed agent, world, and formula,*

$$\mathbf{B}_i^{\text{after}}\varphi = \mathbf{B}_i^{\text{before}}\varphi$$

if and only if the positive threshold decision is unchanged and the negative threshold decision is unchanged. STATUS: LEAN-VERIFIED.

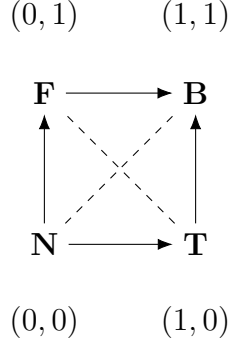
The raw probabilities may therefore move substantially while the categorical result remains fixed. The invariant is not the precise support mass but the pair of threshold sides occupied by the two masses.

The Lean development packages this fact in a compositional predicate, `ModalConditionalizationRobust`. At a belief node, robustness requires only that the two threshold decisions be preserved at every world. The embedded formula itself need not be dynamically invariant. Negation, conjunction, evidence-stable knowledge, and raw possibility propagate robustness compositionally. The previously verified probability-free fragment is contained in this semantic class, but the inclusion is strict: a diagonal member of the reachability family gives a formula containing $\mathbf{B}$ whose probability distribution changes nontrivially while the belief value stays fixed.

5.1 The threshold square

Since every FDE value is already a pair of threshold bits, the categorical state space is the Boolean square

$$\mathbf{N} = (0, 0), \quad \mathbf{T} = (1, 0), \quad \mathbf{F} = (0, 1), \quad \mathbf{B} = (1, 1).$$



Define the threshold-wall count

$$d_{\text{thr}}(a, b) = \mathbf{1}[a^+ \neq b^+] + \mathbf{1}[a^- \neq b^-].$$

Lean verifies the basic geometry:

$$0 \leq d_{\text{thr}}(a, b) \leq 2,$$

$$d_{\text{thr}}(a, b) = 0 \iff a = b,$$

and

$$d_{\text{thr}}(a, b) = 2 \iff a^+ \neq b^+ \wedge a^- \neq b^-.$$

Thus one-wall changes are exactly edge moves and two-wall changes are exactly diagonal moves. The two diagonal pairs are

$$\mathbf{N} \leftrightarrow \mathbf{B}, \quad \mathbf{T} \leftrightarrow \mathbf{F}.$$

Proposition 5.2 (Robustness is zero displacement). *Every compositionally robust modal formula has threshold-wall count zero under its designated admissible conditionalization. STATUS: LEAN-VERIFIED.*

The reachability construction realizes every displacement class. Hence the dynamic picture combines two properties that might initially look opposed:

global categorical reachability is complete,

while

local categorical change is exactly coordinatewise threshold change.

This tension is the starting point for the geometric refinement in the next sections.

6 From Threshold Straddling to Literal Affine Crossings

The Boolean square records which threshold decisions differ, but it does not yet say what numerical event lies behind a changed bit. The next formal layer reconnects categorical displacement to rational support masses.

6.1 Endpoint straddling

For a rational threshold c , define

$$\text{Straddle}(c; x, y)$$

to mean that exactly one endpoint lies on or above the threshold:

$$(c \leq x \wedge \neg(c \leq y)) \vee (c \leq y \wedge \neg(c \leq x)).$$

For rational numbers this is the expected endpoint pattern

$$x < c \leq y \quad \text{or} \quad y < c \leq x.$$

Theorem 6.1 (Threshold decision versus endpoint straddling). *For rational c, x, y ,*

$$[c \leq x] \neq [c \leq y] \iff \text{Straddle}(c; x, y).$$

STATUS: LEAN-VERIFIED.

At belief level, the total wall count decomposes into the one-dimensional wall counts of the positive and negative support coordinates. Lean therefore verifies the exact three-way correspondence:

$$\begin{aligned} d_{\text{thr}} = 0 & \iff \text{neither support coordinate straddles,} \\ d_{\text{thr}} = 1 & \iff \text{exactly one support coordinate straddles,} \\ d_{\text{thr}} = 2 & \iff \text{both support coordinates straddle.} \end{aligned}$$

In particular, every genuine categorical belief change has a numerical endpoint witness on at least one signed support axis.

6.2 Affine interpolation

Endpoint straddling still describes only two snapshots. To expose an actual crossing event, the formalization introduces the directed rational affine path

$$\gamma_{x,y}(t) = x + t(y - x), \quad t \in \mathbb{Q}.$$

It satisfies

$$\gamma_{x,y}(0) = x, \quad \gamma_{x,y}(1) = y.$$

For the increasing orientation $x < c \leq y$, the explicit threshold parameter is

$$t_+ = \frac{c - x}{y - x}.$$

For the decreasing orientation $y < c \leq x$, the development uses the equivalent directed parameter

$$t_- = \frac{x - c}{x - y}.$$

Lean Core rational arithmetic is sufficient to prove that the relevant parameter lies in the unit interval and hits the threshold exactly.

Theorem 6.2 (Affine unit-interval threshold crossing). *If rational endpoints x, y straddle rational threshold c , then*

$$\exists t \in \mathbb{Q} : 0 \leq t \leq 1 \quad \wedge \quad \gamma_{x,y}(t) = c.$$

STATUS: LEAN-VERIFIED.

No abstract topology is used. The proof is constructive for the chosen affine interpolation and introduces no Mathlib dependency.

Corollary 6.3 (Belief change has a literal affine wall hit). *If admissible conditionalization changes the complete FDE value of $B_i\varphi$, then at least one of the two affine support paths*

$$t \mapsto P_t^+(\varphi), \quad t \mapsto P_t^-(\varphi)$$

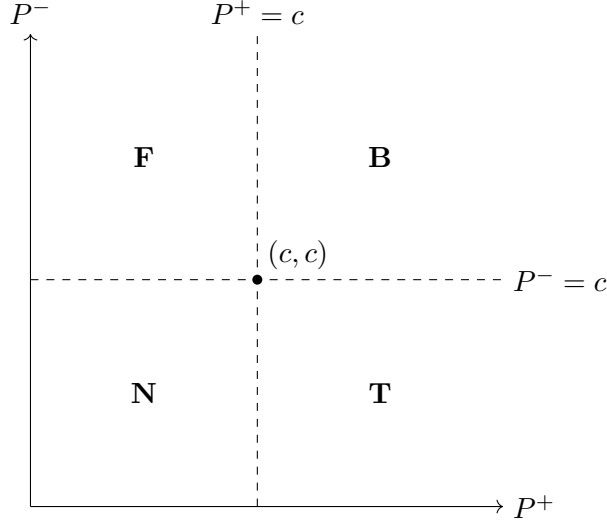
hits c_i at some rational $t \in [0, 1]$. STATUS: LEAN-VERIFIED.

For a two-wall transition, both signed support coordinates have crossing witnesses, though they need not cross at the same parameter. This opens a genuinely temporal question that the endpoint square cannot answer: which wall is hit first?

Support-plane picture. The two masses define a point

$$(P^+, P^-) \in \mathbb{Q}^2.$$

The threshold lines $P^+ = c$ and $P^- = c$ partition the plane into the four FDE regions.



The Boolean threshold square can therefore be read as the coarse four-cell projection of this signed support plane.

7 Unique Crossing Times and Temporal Order

A two-wall transition gives one affine threshold hit on the positive support coordinate and one on the negative coordinate. To turn these hits into a temporal classification, the crossing parameter must be intrinsic rather than an arbitrary existential witness.

7.1 Uniqueness

Suppose $x \neq y$. The affine map

$$\gamma_{x,y}(t) = x + t(y - x)$$

is injective in its parameter. Lean proves this directly over the rationals by cancelling the nonzero factor $y - x$.

Theorem 7.1 (Unique affine threshold time). *Let $x \neq y$. If*

$$\gamma_{x,y}(t) = c \quad \text{and} \quad \gamma_{x,y}(s) = c,$$

then

$$t = s.$$

Consequently every straddling affine segment has a unique threshold crossing parameter in $[0, 1] \cap \mathbb{Q}$.
STATUS: LEAN-VERIFIED.

For a two-wall belief transition write

t_+ = the unique positive-support crossing time,

t_- = the unique negative-support crossing time.

Both lie in the rational unit interval.

7.2 Crossing-order trichotomy

The linear order on $\mathbb{Q}$ now yields an exhaustive classification:

$$t_+ < t_-, \quad t_+ = t_-, \quad t_- < t_+.$$

Theorem 7.2 (Affine crossing-order trichotomy). *Every two-wall conditionalized belief transition admits intrinsic positive and negative crossing times satisfying exactly one of*

$$t_+ < t_-, \quad t_+ = t_-, \quad t_- < t_+.$$

STATUS: LEAN-VERIFIED.

The three cases have immediate event readings:

$$\begin{aligned} t_+ < t_- & : \text{ positive threshold wall first,} \\ t_+ = t_- & : \text{ simultaneous crossing,} \\ t_- < t_+ & : \text{ negative threshold wall first.} \end{aligned}$$

7.3 The simultaneous case

The equality case has a geometric characterization stronger than mere coincidence of two stored parameters.

Theorem 7.3 (Simultaneity as the wall intersection). *For a two-coordinate affine crossing pair,*

$$t_+ = t_-$$

if and only if there exists one rational parameter t such that

$$P_t^+ = c \quad \text{and} \quad P_t^- = c.$$

Equivalently, the support path passes through the wall-intersection point

$$(c, c).$$

STATUS: LEAN-VERIFIED.

Thus simultaneous diagonal motion is geometrically exceptional. If $t_+ \neq t_-$, there is a nonempty rational interval between the two wall events. The next section proves that this interval carries a forced categorical phase.

Why order matters. The endpoint projection forgets which threshold coordinate changes first. Yet for a diagonal transition both bits must change. The temporal order of those changes therefore contains information absent from the pair (source, target). Crossing-order geometry recovers exactly that lost information.

8 Intermediate FDE Phases Between Sequential Crossings

Suppose a two-wall transition is not simultaneous. The two unique crossing times then determine an open interval in which exactly one threshold coordinate has already switched to its target side.

The formal development chooses the rational midpoint

$$t_m = \frac{t_+ + t_-}{2}.$$

If $t_+ < t_-$, Lean proves

$$t_+ < t_m < t_-.$$

If $t_- < t_+$, the symmetric inequalities hold. These midpoint facts require only rational arithmetic.

8.1 Threshold side before and after a crossing

The affine support coordinate is strictly monotone whenever its endpoints are distinct. For an increasing straddling path, parameters before the unique crossing remain below threshold and parameters after it lie above threshold. For a decreasing path, the threshold sides reverse. The formal result can be stated without choosing an orientation:

Proposition 8.1 (Source side before, target side after). *Let a rational affine support coordinate straddle c and hit c uniquely at time t_c . Then*

$$t < t_c \implies \text{side}_c(\gamma(t)) = \text{side}_c(x),$$

and

$$t_c < t \implies \text{side}_c(\gamma(t)) = \text{side}_c(y).$$

STATUS: LEAN-VERIFIED.

This is the key bridge from event order to a categorical intermediate state.

8.2 Generic intermediate vertices

Let

$$s = (s^+, s^-), \quad t = (t^+, t^-)$$

be opposite vertices of the threshold square. Define

$$I_+(s, t) = (t^+, s^-)$$

for positive-first motion and

$$I_-(s, t) = (s^+, t^-)$$

for negative-first motion.

Theorem 8.2 (Positive-first midpoint phase). *If both support coordinates straddle and $t_+ < t_-$, then at the rational midpoint*

$$\text{state}(t_m) = I_+(s, t).$$

That is, the positive coordinate has already reached the target threshold side while the negative coordinate is still on the source threshold side. STATUS: LEAN-VERIFIED.

Theorem 8.3 (Negative-first midpoint phase). *If both support coordinates straddle and $t_- < t_+$, then*

$$\text{state}(t_m) = I_-(s, t).$$

STATUS: LEAN-VERIFIED.

For opposite endpoints, both intermediate states are adjacent to source and target:

$$d_{\text{thr}}(s, I_{\pm}(s, t)) = 1, \quad d_{\text{thr}}(I_{\pm}(s, t), t) = 1.$$

Hence every sequential diagonal displacement decomposes into two one-wall phases.

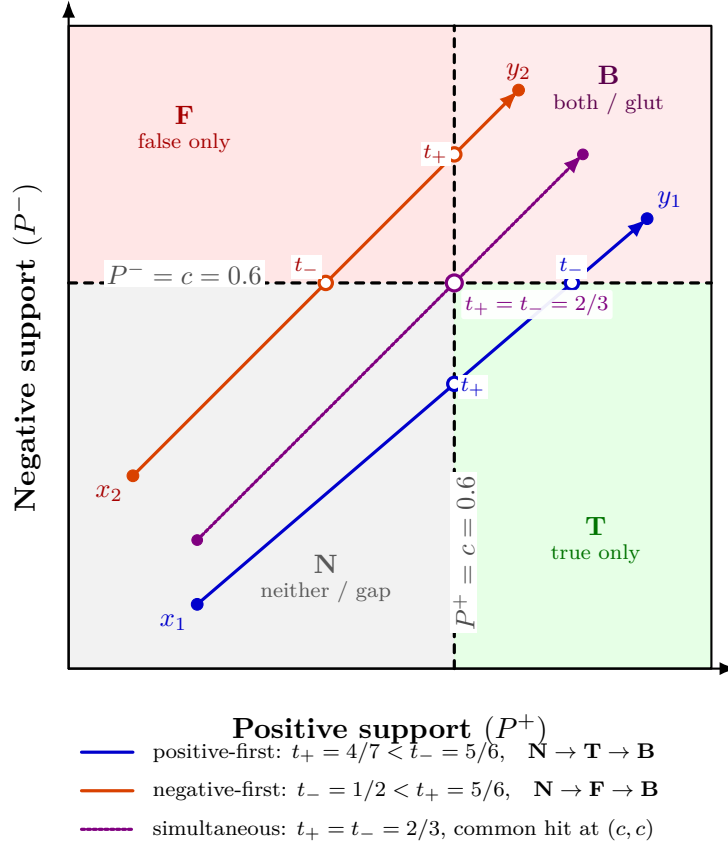


Figure 1: Affine phase geometry for three $\mathbf{N} \rightarrow \mathbf{B}$ support trajectories at threshold $c = 0.6$. Positive-first and negative-first paths necessarily occupy $\mathbf{T}$ and $\mathbf{F}$, respectively, between their unique wall-crossing times. The simultaneous path is the exceptional case $t_+ = t_-$ and passes through the wall intersection (c, c) . The figure visualizes the Lean-verified crossing-order and intermediate-phase theorems; it does not by itself assert that every interpolated support point is already realized by a full admissible model.

8.3 Concrete diagonal table

The complete verified table is:

Diagonal transition	positive wall first	negative wall first
$\mathbf{N} \rightarrow \mathbf{B}$	$\mathbf{T}$	$\mathbf{F}$
$\mathbf{B} \rightarrow \mathbf{N}$	$\mathbf{F}$	$\mathbf{T}$
$\mathbf{T} \rightarrow \mathbf{F}$	$\mathbf{N}$	$\mathbf{B}$
$\mathbf{F} \rightarrow \mathbf{T}$	$\mathbf{B}$	$\mathbf{N}$

Thus, for example,

$$\mathbf{N} \rightarrow \mathbf{B}$$

can have either phase sequence

$$\mathbf{N} \rightarrow \mathbf{T} \rightarrow \mathbf{B}$$

or

$$\mathbf{N} \rightarrow \mathbf{F} \rightarrow \mathbf{B},$$

and the choice is determined exactly by whether positive or negative support crosses the Lockean wall first.

Figure 1 makes the trichotomy explicit for three rational affine trajectories with the same threshold. The two sequential cases pass through different adjacent FDE cells, whereas the simultaneous path hits the unique wall intersection and therefore has no open interval carrying an intermediate adjacent phase.

Theorem 8.4 (Two-wall intermediate-phase classification). *Every two-wall conditionalized belief transition admits an affine crossing pair such that either*

- (i) *the two crossing times are equal and the support path hits (c, c) , or*
- (ii) *the positive crossing is earlier and the midpoint has the positive-first intermediate FDE state, or*
- (iii) *the negative crossing is earlier and the midpoint has the negative-first intermediate FDE state.*

STATUS: LEAN-VERIFIED.

Scope of this theorem. The theorem in this section classifies the explicit affine interpolation of the two support masses. Section 9 subsequently verifies a genuine path of complete strong 4-PEL models for weight-generated endpoints and proves that every fixed event mass follows exactly such an affine trajectory. The remaining gap is therefore no longer model existence in general, but the formula-level identification of the positive and negative support events with fixed events along the model path. For arbitrary probability-sensitive modal formulas those event memberships may themselves change with the path parameter.

9 From Probability Simplexes to Strong Model Paths

The affine crossing results of the previous sections are formulated at the level of signed support masses. To interpret those trajectories as genuine probabilistic model trajectories, the probability layer must first satisfy more than the lightweight normalization contract built into the original `Model` structure. This section records the stronger layer and the verified lift from rational weight vectors to complete 4-PEL models.

9.1 Finite probability integrity

The compatibility predicate `FiniteProbabilityIntegrity` strengthens one local measure by requiring duplicate-free support, nonnegative event masses, extensionality with respect to event membership, monotonicity, disjoint finite additivity, empty-event mass zero, and total support mass one. The wrapper `StrongProbabilityModel` equips an ordinary 4-PEL model with this stronger certification.

The stronger contract is deliberately additive rather than a replacement of the legacy model interface. Earlier finite witnesses therefore remain available, while later geometric results can state explicitly when genuine finite-probability behavior is required.

Proposition 9.1 (Weight-generated finite probability). *Let R be a duplicate-free finite support and let $q : R \rightarrow \mathbb{Q}$ be nonnegative with total mass one. Define the mass of a finite event S by summing the weights of the points of R that belong to S . Then the resulting local measure satisfies `FiniteProbabilityIntegrity`. STATUS: LEAN-VERIFIED.*

The implementation iterates over the fixed support rather than over the event list. Consequently event order and duplicate presentation do not alter the generated mass.

9.2 Convexity of the finite probability simplex

For two valid weight distributions q_0, q_1 on the same support and rational $t \in [0, 1]$, define

$$q_t(x) = (1 - t)q_0(x) + tq_1(x).$$

Lean verifies that q_t is again a valid finite weight distribution. Hence the full rational line segment between two valid endpoints remains in the finite probability simplex.

More strongly, every fixed event mass is affine in the same parameter:

$$\mu_t(S) = (1 - t)\mu_0(S) + t\mu_1(S).$$

Theorem 9.2 (Affine event mass on the probability simplex). *For every fixed event S , the weight-generated mass along the convex distribution path satisfies*

$$\mu_{q_t}(S) = (1 - t)\mu_{q_0}(S) + t\mu_{q_1}(S).$$

STATUS: LEAN-VERIFIED.

This identity is the formal bridge between the probability simplex and the affine scalar paths used in the threshold-crossing theorems.

9.3 A model-valued rational path

The probability path can be lifted to the complete model structure. Fix a world list W , an accessibility field R , an atomic valuation v , and Lockean thresholds c . Let q_0 and q_1 provide valid local weight distributions at every agent/world pair. For every rational $t \in [0, 1]$, define

$$M_t = (W, R, \mu_t, v, c),$$

where each local measure is generated from the interpolated weights q_t .

Theorem 9.3 (Convex strong model path). *Every rational $t \in [0, 1]$ determines a **StrongProbabilityModel** M_t . Along the entire path,*

$$W_t = W, \quad R_t = R, \quad v_t = v, \quad c_t = c,$$

and every fixed local event has affine probability mass. STATUS: LEAN-VERIFIED.

Thus the geometric probability path is realized by complete probabilistically certified 4-PEL models, rather than merely by pairs of numbers in a support plane.

Exact scope of the realization. Two distinctions remain essential. First, this theorem concerns weight-generated strong endpoint models on a common semantic skeleton. It does not retroactively prove that every legacy witness in the repository satisfies the stronger probability contract. Second, a convex model path is not the same thing as an update-generated path: the theorem does not claim that every intermediate M_t arises by conditionalizing the source model on some evidence event.

There is also a formula-level boundary. The affine theorem applies directly to a *fixed event* S . For an arbitrary modal formula containing probabilistic belief, the set of worlds that positively or negatively support the formula can itself vary with t . In that case the event whose mass is measured is no longer fixed, so affine event mass alone does not imply affine `modalPositiveBeliefMass` or `modalNegativeBeliefMass`.

The next formal step is therefore narrower and sharper: identify a path-invariant formula fragment, beginning with atomic propositions and probability-free formulas, and lift the verified threshold-crossing and intermediate-phase geometry to those genuine model paths.

10 Structural Interpretation: From Reachability to Phase Kinematics

STATUS: STRUCTURAL INTERPRETATION.

The formal results support a layered interpretation of four-valued epistemic change. At the finest level considered here, a belief formula is represented by two rational support masses. Thresholding projects this point into one of four regions. The resulting FDE value forgets the precise probabilities and remembers only which side of the two Lockean walls each coordinate occupies.

This can be written as a coarse-graining map

$$q_c : \mathbb{Q}^2 \longrightarrow \{\mathbf{T}, \mathbf{F}, \mathbf{B}, \mathbf{N}\},$$

$$q_c(p^+, p^-) = ([p^+ \geq c], [p^- \geq c]).$$

The two threshold lines partition the support plane into four cells. The Boolean threshold square is the adjacency graph of these cells. In this sense the four-valued state space is not an arbitrary collection of labels. It is the quotient of a signed evidence space by two threshold predicates.

10.1 Global freedom, local constraint

Two verified results together give the central dynamic contrast:

every ordered categorical transition is realizable,

while

every categorical change is mediated by a threshold-side change.

Global reachability is therefore maximal, but local motion is not structureless. A transition leaves a trace in the signed support coordinates. Zero-wall movement is robust, one-wall movement crosses one categorical boundary, and two-wall movement crosses both.

The affine results sharpen this further. A diagonal transition does not merely differ in both bits. Along the chosen support interpolation it contains two unique wall events. Those events may coincide, or they may be temporally ordered. If they are ordered, a forced adjacent phase lies between them. The endpoint displacement therefore hides a short event history.

It is useful to call this structure *epistemic phase kinematics*. The term is not an additional theorem. It names the combination of four verified layers:

cell membership, wall count, crossing time, intermediate phase sequence.

10.2 The exceptional role of simultaneity

For a diagonal move, sequential crossings force an intermediate adjacent phase. Simultaneity is the unique way in which this ordered two-step decomposition disappears. Geometrically, simultaneity requires the affine support path to pass through the codimension-two intersection (c, c) . It is therefore a special alignment condition in the support plane rather than the generic consequence of changing both bits.

This distinction may become important in later model-level dynamics. If admissible model paths can realize the affine support interpolation, sequential diagonal changes would carry genuine intermediate epistemic states, while simultaneous changes would behave as boundary-coincident transitions.

10.3 Connection to structural transport

A broader working hypothesis in the 4-PEL project treats epistemic paradoxes as failures of structural transport. The present dynamic results fit that program in a precise but limited way. Thresholding is a projection from rich signed probabilistic information to a coarse four-valued state. Endpoint states alone do not preserve crossing order. Recovering the order requires returning to the richer support trajectory.

The paper does not claim that all epistemic paradoxes reduce to threshold crossing. Rather, this case study supplies one clean instance of a recurring pattern:

rich structure $\longrightarrow$ coarse projection $\longrightarrow$ apparent categorical jump,

where restoring the discarded structure reveals the mechanism hidden by the projection.

The formal contribution of the present paper ends before this philosophical generalization. A general structural-transport theorem connecting threshold dynamics to the project's other paradox families remains open.

11 Lean Mechanization and Verification Status

The results in this manuscript are implemented in Lean 4.31. The current extension is on branch **research/modal-probabilistic-fine-graining**, building on the earlier dynamic research line. The root module `PEL4.lean` imports the development, so a successful

```
lake build
```

checks the theorem chain together with the rest of the project.

The implementation deliberately uses small semantic layers. The principal files for the present manuscript are:

Lean module	role in the paper
PEL4/FDE.lean	four bilateral values and basic operations
PEL4/Model.lean	finite probabilistic Kripke model interface
PEL4/FiniteProbabilityIntegrity.lean	stronger finite-probability compatibility layer
PEL4/WeightGeneratedProbability.lean	weight-generated measures and generic integrity theorem
PEL4/ConvexProbabilitySimplex.lean	convex rational distributions and affine event masses
PEL4/ConvexModelPath.lean	complete strong model-valued convex paths
PEL4/Belief.lean	signed Lockean threshold belief
PEL4/ModalLanguage.lean	primitive B, K, and raw possibility
PEL4/ModalKnowledgeBeliefFactorization.lean	exact $K = \text{Stable}$ -filtered B theorem
PEL4/Dynamics.lean	admissible probability-only conditionalization
PEL4/ModalDynamicsStability.lean	probability-free invariance
PEL4/ModalDynamicsPhaseClassification.lean	four stability phases for knowledge
PEL4/ModalDynamicsReachability.lean	complete sixteen-transition witness family
PEL4/ModalDynamicsRobustness.lean	exact threshold-side robustness boundary
PEL4/ModalDynamicsGeometry.lean	Boolean threshold square and wall count
PEL4/ModalDynamicsThresholdCrossing.lean	numerical endpoint straddling
PEL4/ModalDynamicsAffineCrossing.lean	explicit rational affine threshold hits
PEL4/ModalDynamicsCrossingOrder.lean	unique crossing times and order trichotomy
PEL4/ModalDynamicsIntermediatePhase.lean	forced midpoint phase between sequential crossings

11.1 Proof style

The development combines generic theorems with finite executable witnesses. The generic parts include the knowledge/belief factorization, robustness equivalence, Boolean wall-count laws, rational affine crossing, crossing uniqueness, order trichotomy, midpoint phase classification, finite-probability integrity, convexity of the rational probability simplex, affine fixed-event mass, and construction of complete strong model paths. Finite constructions are used where an existential semantic possibility is at issue, most notably for stability fracture/restoration and complete dynamic reachability.

Several inherited finite cases use `native_decide`, which relies on native evaluation and has a different trust boundary from kernel reduction alone. The new MPFG gate instead uses `decide+kernel` for its rational witnesses. Its focused audit checks all 24 selected declarations and rejects project-specific axioms, `sorryAx`, and native-decision axioms; only `propext`, `Classical.choice`, and `Quot.sound` are permitted. This audit is not a claim about every legacy declaration in the repository. Finite witnesses and general symbolic theorems also remain distinct in their quantifier structure.

The affine path development intentionally avoids adding Mathlib. It imports Lean Core rational lemmas and proves the required unit-interval, linearity, and monotonicity facts directly. This keeps the current result lightweight and makes the boundary to a future topology-heavy

development explicit.

11.2 Status discipline

The manuscript uses the following hierarchy:

- (1) **Lean-verified theorem:** a universally quantified or closed proposition compiled successfully through the active root import.
- (2) **Finite executable witness:** a concrete model theorem checked by computation inside Lean.
- (3) **Structural interpretation:** a conceptual reading of the formal results, not a separately mechanized theorem.
- (4) **Literature or novelty claim:** a comparative scholarly assertion requiring an external audit.

The central dynamic and probability-path modules through `ConvexModelPath.lean` have passed fresh local `lakebuild` checks. Accordingly, the reachability, robustness, wall geometry, affine crossing, crossing-order, intermediate-phase, finite-probability integrity, probability-simplex, and strong model-path claims reported in this manuscript are machine-checked at the level of their exact Lean statements.

Reproducibility. A minimal reproduction sequence is:

```
git fetch origin
git switch research/modal-probabilistic-fine-graining
git pull --ff-only
lake build
python3 scripts/check_mpfg_axioms.py
```

The paper source itself is intentionally separate from the Lean build so that manuscript compilation does not become a prerequisite for checking the formal logic.

12 Related Work, Limits, and Open Problems

The semantic kernel belongs to the Belnap–Dunn/FDE tradition of tracking support for truth and falsity independently [1, 2]. Four-valued modal logics with bilateral relational semantics have been studied in several forms [4, 5]. The present project adds a finite probabilistic threshold layer and a primitive evidence-stable knowledge operator whose behavior is analyzed mechanically rather than assumed to coincide with a standard modal box.

The threshold connection between all-or-nothing belief and degrees of belief belongs to the broader Lockean tradition. Leitgeb’s stability theory is a particularly important point of comparison because it places stability conditions at the interface between probability and categorical belief [3]. The 4-PEL notion of stability is different in type: it concerns invariance of the complete FDE value across an accessible profile and acts as a filter from threshold belief to knowledge. A careful conceptual comparison is therefore needed before stronger claims about similarity or difference are made.

The update perspective is adjacent to dynamic epistemic logic, where informational actions transform epistemic models and knowledge states [6]. The conditionalization operator studied here is narrower: it preserves worlds, accessibility, valuation, and threshold and changes only local probabilities. Conversely, the four-valued signed-support geometry developed here is not part of the standard DEL vocabulary. The manuscript reports this construction without claiming that it is novel relative to all probabilistic or nonclassical dynamic epistemic systems.

Novelty boundary. No strong novelty claim is made in this version. The individual mathematical ingredients are elementary: Boolean pairs, threshold comparisons, affine interpolation, finite convex probability, linear order, and midpoint arguments. The research contribution currently established internally to the project is the machine-checked composition of these ingredients into one dependency chain connecting admissible probabilistic update to four-valued categorical reachability, robustness, crossing order, intermediate phase structure, and a model-valued rational probability path. A systematic literature audit is publication-critical.

12.1 Verified model-path layer and remaining realization questions

The earlier support-to-model existence problem now has a positive answer for a precise class. Given two weight-generated strong endpoint models on the same semantic skeleton, rational convex interpolation of their local weights produces a **StrongProbabilityModel** for every $t \in [0, 1] \cap \mathbb{Q}$. Worlds, accessibility, valuation, and threshold remain fixed, and the probability of every fixed event varies affinely.

Two stronger realization questions remain open.

Research Question 12.1 (Formula-level support realization). For which modal formulas φ are the positive and negative support events invariant along a convex strong model path, so that the verified affine fixed-event theorem applies directly to $P_t^+(\varphi)$ and $P_t^-(\varphi)$?

Atomic propositions are the natural first case because their valuation is fixed by construction. A broader probability-free fragment is plausible but requires an explicit theorem connecting path-invariance of **evalModal** to fixed support-event membership.

Research Question 12.2 (Update-generated path). When can the intermediate models of a convex strong model path themselves be obtained by admissible conditionalization of the source model, possibly using a parameterized family of evidence events?

A model-valued path and an update-generated path are different notions. C17D verifies the former and deliberately does not infer the latter.

12.2 Open problem 2: arbitrary continuous paths

The affine crossing theorem is constructive and sufficient for the present phase result, but it is not a general topological theorem. A stronger layer would define a continuous support path

$$\gamma : [0, 1] \rightarrow \mathbb{R}^2$$

and prove that endpoint straddling forces intersection with the appropriate threshold wall. Standard intermediate-value reasoning would then recover a crossing theorem independent of linear interpolation.

For two-wall transitions, arbitrary paths introduce genuinely new geometry. A path may meet the two walls multiple times, touch a wall without changing side, or revisit a categorical cell. The current uniqueness and three-case temporal order are specific to nonconstant affine coordinates. A general theory would therefore need a richer notion of crossing event than the present unique hit parameter.

12.3 Open problem 3: algebraic stability

The knowledge filter is presently defined semantically by full-value homogeneity over an accessible range. An open problem is to characterize this stability intrinsically in bilattice or algebraic terms and to determine which logical operators preserve, reflect, or destroy it. Such a characterization could connect the dynamic phase table to existing algebraic semantics for four-valued modal systems.

12.4 Open problem 4: higher-dimensional phase complexes

A single belief formula contributes two threshold bits. For n tracked formulas, the naive categorical state space has $2n$ threshold coordinates and therefore resembles a Boolean hypercube. This suggests a higher-dimensional generalization of wall count, crossing order, and phase sequence. Dependencies between formulas, however, can identify or forbid regions of that cube. Determining the realizable cell complex is a natural extension of the present two-dimensional result.

12.5 Open problem 5: structural transport and paradox families

The project contains independent formal diagnoses of Lottery, Preface, Sorites, Knower, Surprise Examination, Moore/Fitch, and dynamic threshold phenomena. The working phrase “paradoxes as failures of structural transport” is currently interpretive. A genuine general theorem would require an abstract account of transformations, observations, preserved predicates, and counterexamples that subsumes several of these families without erasing their differences.

13 A conservative modal-probabilistic refinement gate

This extension is a semantic refinement of evidence, not a replacement of the four-valued algebra. Its formal declarations are in `PEL4/ModalProbability/Refinement.lean` and `PEL4/ModalProbability/StatusTransport.lean`. The gate’s validation status is recorded in `docs/MODAL_PROBABILISTIC_FINE_GRAINING.md`.

Definition 13.1 (Reliability-tagged evidence profile). A fine profile is a nonnegative rational vector $q = (t, b, n, f, t_r, f_r)$ of total mass one. Its coarse projection is

$$\pi(q) = (t + t_r, b, n, f + f_r).$$

The reliability tags are supplied semantic data, not proofs of empirical truth. The untagged lift of (t, b, n, f) is $(t, b, n, f, 0, 0)$.

Consequently π has a right inverse. Signed support is preserved: $P^+(q) = t + t_r + b$ and $P^-(q) = f + f_r + b$. For $0 \leq a \leq 1$, both normalized rational simplexes are closed under mixing, and the formal projection identity is

$$\pi((1 - a)q + ar) = (1 - a)\pi(q) + a\pi(r).$$

Thus raw threshold belief commutes with forgetting reliability. This does not assert that thresholding is affine; the raw belief compatibility follows from the chosen definitions. Moreover, no function of the coarse profile recovers reliable mass $t_r + f_r$ in general: pure untagged truth and pure reliable truth have the same projection, but reliable masses zero and one.

Definition 13.2 (Status stability and local fibre rigidity). For any relation R , observation v , and projection π , put

$$\begin{aligned} \text{Stable}(v, w) &\iff \forall u (Rwu \Rightarrow v(u) = v(w)), \\ \text{Rigid}(v, \pi, w) &\iff \forall u (Rwu \Rightarrow (\pi(v(u)) = \pi(v(w)) \Rightarrow v(u) = v(w))). \end{aligned}$$

The exact transport boundary is

$$\text{Stable}(v, w) \iff \text{Stable}(\pi \circ v, w) \wedge \text{Rigid}(v, \pi, w).$$

This is a local decomposition, not a claim that fibre rigidity is automatic. Two universally accessible worlds carrying untagged and reliable truth witness coarse stability with fine instability, even on a reflexive, transitive, Euclidean frame. A second witness distinguishes mass stability

from categorical stability: $(0, 1, 0, 0)$ and $(0, 3/4, 1/4, 0)$ both yield **B** at threshold $3/5$, although their glut masses differ.

The auxiliary essence predicate means conditionally retaining the current specified status across accessible worlds; it may hold vacuously. Its accidental counterpart requires an actual different accessible status. These are classical meta-level predicates, not additional FDE connectives. Their relationship to the existing executable stability test and **K** is stated separately: applying the unchanged knowledge interpretation to raw fine belief or to its coarse projection yields the same value. No identification of **K** with ordinary S5 necessity is made.

Finally, the existing finite-step-chain theorem transports threshold status when each edge preserves it. Cell refinement is distinct from population Finite Fine-Grainedness, and connectivity alone does not imply status preservation. Normalized reliability-sensitive updates, probability-one versus necessity under full-support assumptions, and any representation of proof-theoretic fractional values remain subsequent research targets.

14 Conclusion

The dynamic 4-PEL development begins with a simple threshold semantics and ends with a surprisingly structured phase picture. Positive and negative evidence are tracked independently, so probabilistic belief is a pair of Lockean threshold decisions. Evidence-stable knowledge is exactly that belief value filtered by qualitative homogeneity of the complete accessible FDE profile.

Admissible conditionalization changes only probability, yet its categorical effects are maximally expressive. A six-world family realizes every ordered transition between **T, F, B, N** at the level of $K(\mathbf{B}p)$. Gluts and gaps can both be created and removed. This global reachability does not make the dynamics arbitrary. Pointwise belief invariance is characterized exactly by preservation of the two threshold bits.

Those bits supply a natural Boolean geometry. The four FDE values form a threshold square, and the number of changed coordinates gives a wall count of zero, one, or two. Numerical endpoint analysis then identifies a changed bit with straddling of the corresponding support threshold. The affine interpolation of rational support masses turns straddling into a literal threshold-hit theorem with an explicit parameter in the unit interval.

For diagonal two-wall transitions, the two support coordinates have unique crossing times. Their order adds temporal structure absent from the endpoints. Simultaneous crossing is exactly passage through (c, c) . Sequential crossing forces an adjacent intermediate phase at the rational midpoint. Consequently, a diagonal endpoint transition has a short hidden phase history. For $\mathbf{N} \rightarrow \mathbf{B}$, positive-first motion yields $\mathbf{N} \rightarrow \mathbf{T} \rightarrow \mathbf{B}$, while negative-first motion yields $\mathbf{N} \rightarrow \mathbf{F} \rightarrow \mathbf{B}$. The remaining diagonals obey the corresponding symmetric table.

The probability layer can now also be given a stronger constructive foundation. Nonnegative normalized rational world weights generate finite measures satisfying explicit integrity laws. The rational segment between two such distributions remains inside the finite probability simplex, and the mass of every fixed event moves affinely. These interpolated local distributions can be packaged into complete **StrongProbabilityModel** paths while worlds, accessibility, valuation, and threshold remain fixed.

The main verified dependency chain is therefore

$ \begin{aligned} &\text{finite probability integrity} \rightarrow \text{probability simplex} \\ &\rightarrow \text{strong model path} \rightarrow \text{affine fixed-event mass} \\ &\rightarrow \text{threshold side} \rightarrow \text{categorical wall} \rightarrow \text{crossing order} \\ &\rightarrow \text{intermediate FDE phase.} \end{aligned} $
--

This chain turns the four-valued endpoint space into more than a classification table. It gives

the beginning of a kinematics of epistemic phases in which a genuine model-valued probability path can project to a piecewise categorical FDE history.

The remaining lifting problem is now sharper than before. Model existence along the rational convex path is verified for weight-generated strong endpoints, but the positive and negative support events of an arbitrary probability-sensitive modal formula need not remain fixed as the model varies. The next step is therefore to characterize a path-invariant formula fragment, beginning with atomic and probability-free formulas, and to show that its signed belief-support masses inherit the verified affine event-mass theorem. A separate question asks when the intermediate models can themselves be generated by admissible conditionalization rather than merely connected by a valid convex probability path.

Only after those formula-level and update-level distinctions are settled should the development move to arbitrary continuous paths or general measure spaces. The finite rational theory already establishes complete reachability, exact robustness, wall geometry, event order, forced intermediate phases, finite probability integrity, and strong model-valued convex paths, while keeping stronger topological and literature claims explicit rather than hidden inside the formal results.

A Formal Correspondence

This appendix maps the principal mathematical claims in the manuscript to the Lean modules and theorem names that support them. The list is selective rather than exhaustive.

Paper claim	Lean theorem / definition	Module
Four FDE values as bilateral Boolean pairs	FDEValue, FDEValue.T/F/B/N	PEL4/FDE.lean
Signed threshold belief	belief	PEL4/Belief.lean
Primitive modal evaluator with $B, K, \Diamond$	ModalFormula, evalModal	PEL4/ModalLanguage.lean
Knowledge equals stability-filtered belief	modal_knowledge_as_stability_filtered_belief	PEL4/ModalKnowledgeBeliefFactorization.lean
Safe probability-only update	ConditionalizationAdmissible, conditionalize	PEL4/Dynamics.lean
Probability-free invariance	evalModal_conditionalize_of_probabilityFree	PEL4/ModalDynamicsStability.lean
Four dynamic stability phases	modal_knowledge_phase_*	PEL4/ModalDynamicsPhaseClassification.lean
Both stability-change directions are realizable	conditionalization_realizes_both_stability_phase_directions	PEL4/ModalDynamicsPhaseClassification.lean
All sixteen $K(Bp)$ endpoint transitions	conditionalization_realizes_every_knowledge_value_transition	PEL4/ModalDynamicsReachability.lean
Creation of a knowledge glut	conditionalization_can_create_knowledge_glut	PEL4/ModalDynamicsReachability.lean
Glut-to-gap and gap-to-truth transitions	conditionalization_can_turn_knowledge_glut_into_gap, ..gap_into_truth	PEL4/ModalDynamicsReachability.lean
Exact belief robustness boundary	modal_belief_value_eq_iff_threshold_bits_eq	PEL4/ModalDynamicsRobustness.lean

Compositional robust formulas are invariant	<code>evalModal_conditionalize_of_robust</code>	<code>PEL4/ModalDynamicsRobustness.lean</code>
Robustness strictly extends the probability-free fragment	<code>diagonal_reachability_belief_is_robust, ...not_probabilityFree</code>	<code>PEL4/ModalDynamicsRobustness.lean</code>
Threshold wall count	<code>thresholdWallCount</code>	<code>PEL4/ModalDynamicsGeometry.lean</code>
Zero wall iff equal FDE state	<code>thresholdWallCount_eq_zero_iff</code>	<code>PEL4/ModalDynamicsGeometry.lean</code>
Two walls iff both coordinates flip	<code>thresholdWallCount_eq_two_iff</code>	<code>PEL4/ModalDynamicsGeometry.lean</code>
Wall flip iff endpoint straddling	<code>threshold_decision_ne_iff_straddles</code>	<code>PEL4/ModalDynamicsThresholdCrossing.lean</code>
Two walls iff both supports straddle	<code>conditionalization_belief_two_walls_iff_both_supports_straddle</code>	<code>PEL4/ModalDynamicsThresholdCrossing.lean</code>
Affine path	<code>affineRatPath</code>	<code>PEL4/ModalDynamicsAffineCrossing.lean</code>
Straddling gives unit-interval threshold hit	<code>thresholdStraddles_has_affine_unit_crossing</code>	<code>PEL4/ModalDynamicsAffineCrossing.lean</code>
Every belief change has an affine wall hit	<code>conditionalization_belief_change_has_affine_threshold_crossing</code>	<code>PEL4/ModalDynamicsAffineCrossing.lean</code>
Two-wall transition has two affine hits	<code>conditionalization_belief_two_walls_has_two_affine_crossings</code>	<code>PEL4/ModalDynamicsAffineCrossing.lean</code>
Unique affine threshold time	<code>affineRatPath_threshold_hit_unique</code>	<code>PEL4/ModalDynamicsCrossingOrder.lean</code>
Crossing-order trichotomy	<code>affineCrossingPair_order_trichotomy</code>	<code>PEL4/ModalDynamicsCrossingOrder.lean</code>
Simultaneous iff common (c, c) hit	<code>affineCrossingPair_simultaneous_iff_common_hit</code>	<code>PEL4/ModalDynamicsCrossingOrder.lean</code>
Conditionalized two-wall order classification	<code>conditionalization_belief_two_walls_crossing_order</code>	<code>PEL4/ModalDynamicsCrossingOrder.lean</code>
Midpoint lies between sequential crossings	<code>affineCrossingMidpoint_between_*</code>	<code>PEL4/ModalDynamicsIntermediatePhase.lean</code>
Source side before and target side after crossing	<code>affine_threshold_side_before_crossing_eq_source, ...after...target</code>	<code>PEL4/ModalDynamicsIntermediatePhase.lean</code>
Positive-first midpoint state	<code>affineCrossingPair_positive_first_midpoint_state</code>	<code>PEL4/ModalDynamicsIntermediatePhase.lean</code>
Negative-first midpoint state	<code>affineCrossingPair_negative_first_midpoint_state</code>	<code>PEL4/ModalDynamicsIntermediatePhase.lean</code>
Intermediate is adjacent to both diagonal endpoints	<code>positiveFirstIntermediate_adjacent_to_diagonal_endpoints, negativeFirstIntermediate_adjacent_to_diagonal_endpoints</code>	<code>PEL4/ModalDynamicsIntermediatePhase.lean</code>

Concrete eight-case diagonal table	<code>affine_diagonal_intermediate_vertex_table</code>	PEL4/ModalDynamicsIntermediatePhase.lean
Full simultaneous/sequential intermediate-phase theorem	<code>conditionalization_belief_two_walls_intermediate_phase</code>	PEL4/ModalDynamicsIntermediatePhase.lean
Strong finite-probability contract	<code>FiniteProbabilityIntegrity, StrongProbabilityModel</code>	PEL4/FiniteProbabilityIntegrity.lean
Weight-generated measure satisfies integrity	<code>weightGeneratedMeasure_integrity</code>	PEL4/WeightGeneratedProbability.lean
Convex interpolation preserves distributions	<code>convexWeightDistribution</code>	PEL4/ConvexProbabilitySimplex.lean
Every fixed event mass is affine	<code>weightedEventMass_convex</code>	PEL4/ConvexProbabilitySimplex.lean
Complete strong rational model path	<code>convexStrongModelAt</code>	PEL4/ConvexModelPath.lean
Worlds, accessibility, valuation, and threshold fixed	<code>convexStrongModelAt_worlds, ...accessibility, ...valuation, ...threshold</code>	PEL4/ConvexModelPath.lean
Fixed-event mass affine along full model path	<code>convexStrongModelAt_eventMass</code>	PEL4/ConvexModelPath.lean

Verification note. The active branch has passed fresh local Lean 4.31 builds through `PEL4/ConvexModelPath.lean`. The manuscript therefore treats the claims listed above as machine-checked at the formal level represented by their exact theorem statements. The formula-level lift from fixed events to arbitrary probability-sensitive support events remains separate and is not inferred from the model-path theorem.

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